LLM EVALUATION · AGENTS

Two LLM judges agreed 92% of the time.

Then I made the questions harder. Agreement collapsed to 61%.

swipe →

01 · THE CONTEXT

LLM-as-judge is everywhere now.

But agreement is not correctness.
Here's a stats-native way to tell them apart.

THE METHOD

Treat judge agreement as **inter-rater reliability**: Cohen's κ with bootstrapped CIs.

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an AP agent + a judge-calibration harness

02 · THE EASY SET

Right-vs-wrong-number questions.

COHEN'S K (N = 26)

0.92

Near-perfect agreement. Looks great. Ship it?

03 · THE HARD SET

Does the judge prefer a confident hallucination over an honest "no data"?

ADD 10 HONESTY-PROBING ITEMS → K (N = 36)

0.61

A **31-point** collapse from one design change.

04 · THE REVEAL

Size doesn't predict judgment.

JUDGE (ON THE HARD SUBSET)	ACCURACY
llama-3.3-70b	**20%**
gpt-oss-20b (≈3.5× smaller)	**70%**

The big model chose the fabricated answer over "no data" on **5 of 5** unknown-vendor questions.

05 · THE BIAS

A portable failure mode.

LLM judges systematically reward **confident specificity** over **honest uncertainty** — across labs and model sizes.

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the judges even say so in their own reasoning

06 · WHY IT MATTERS

The wrong judge trains the wrong agent.

An agent that invents a vendor total is **worse** than one that says "no data."

A judge that rewards the invention optimizes your agents toward exactly that.

TAKEAWAY

- High agreement on easy items is not evidence of judge quality.
- Bigger isn't a better judge.
- Probe epistemic honesty, or your eval is theater.

Follow for more on **evaluating AI honestly.**

ap-agent

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